GPUs
CUDA

Vladislav Goncharenko
Senior researcher

girafe ai

MIPT, spring 2025

Recap

Highload

- Foundations
  - von Neumann architecture
  - Flynn’s taxonomy
  - Data locality principle
- MapReduce paradigm
  - Hadoop stack
  - YTsaurus
- kNN indexes

Outline

GPUs, CUDA

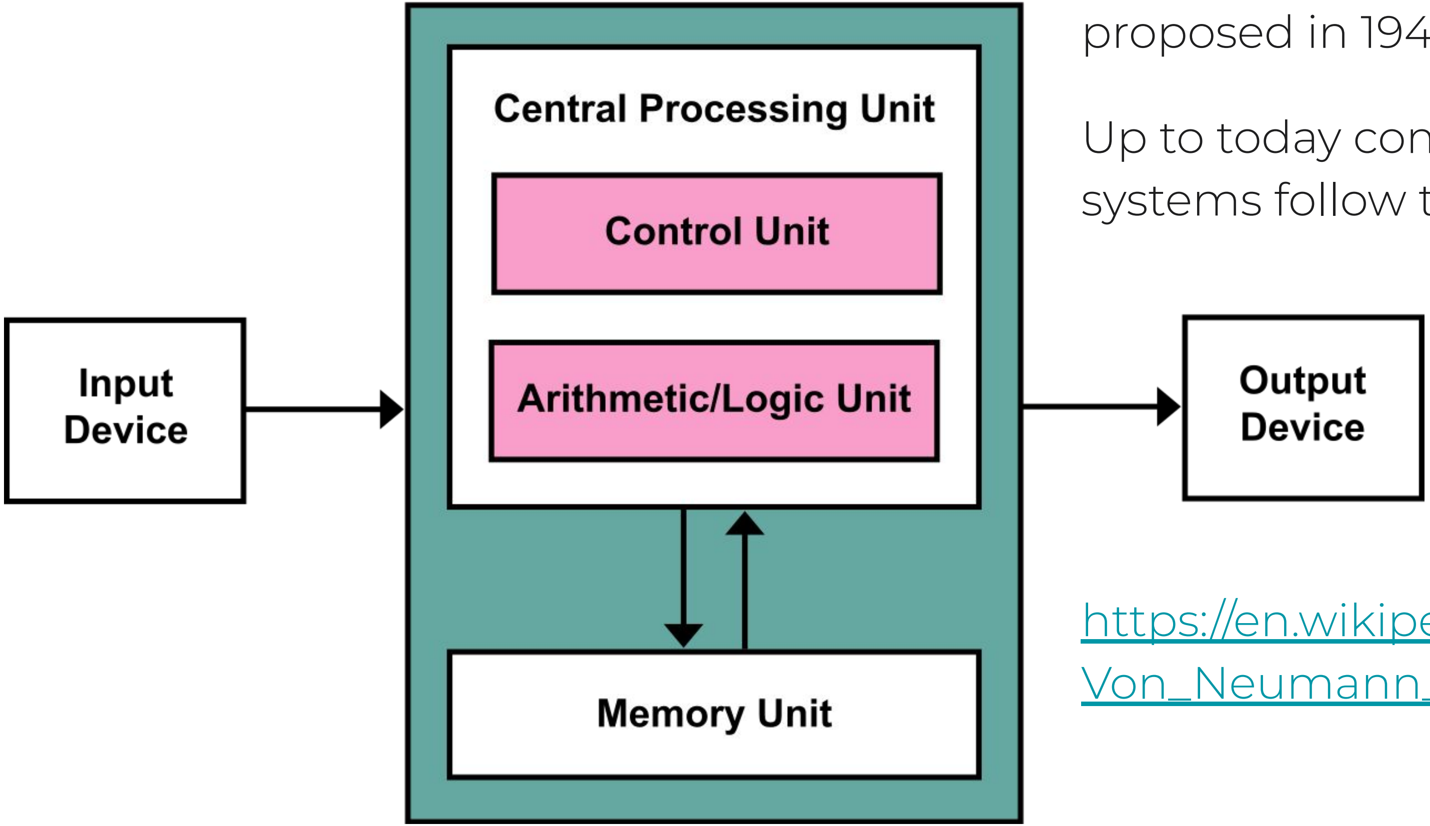
•

Hardware

girafe
ai

01

von Neumann architecture

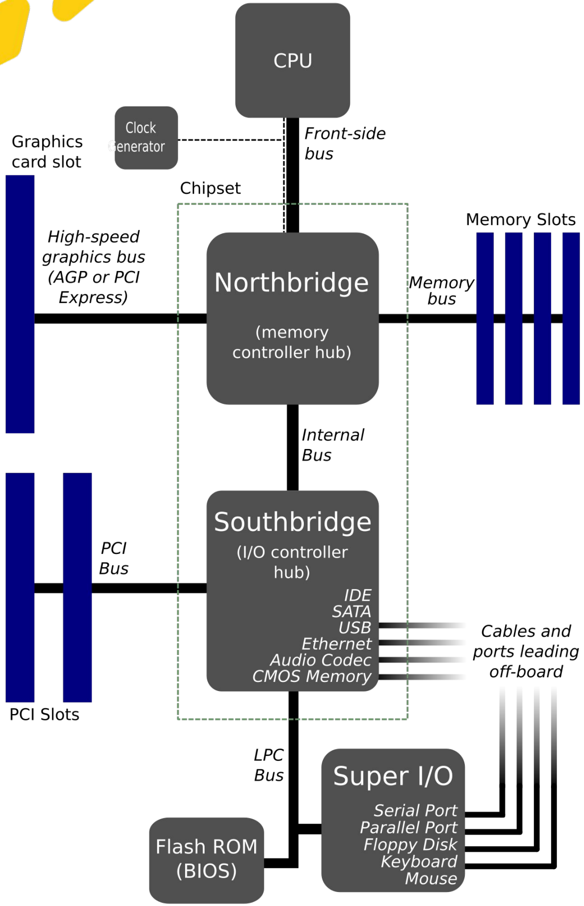


Computer architecture proposed in 1945.

Up to today computer systems follow this design.

https://en.wikipedia.org/wiki/Von_Neumann_architecture

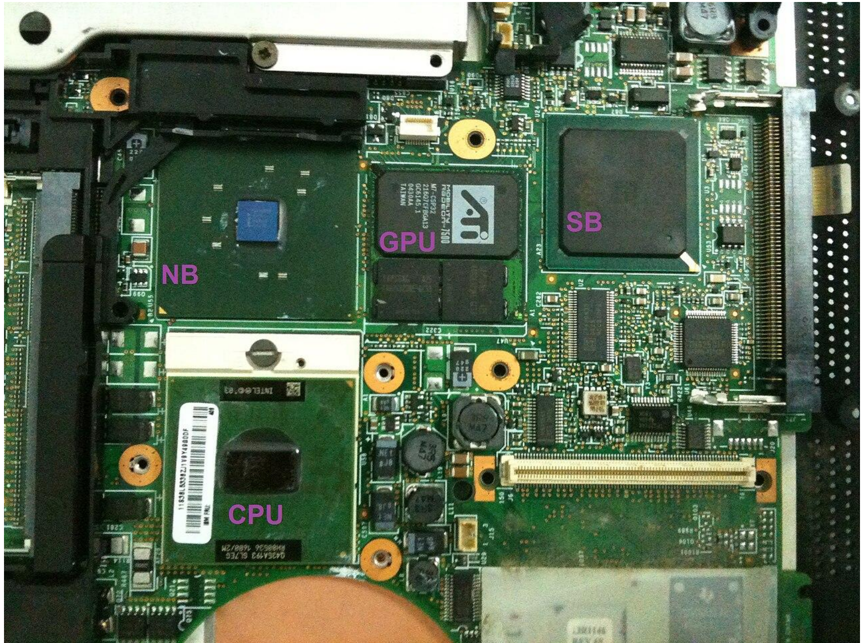
von Neumann architecture



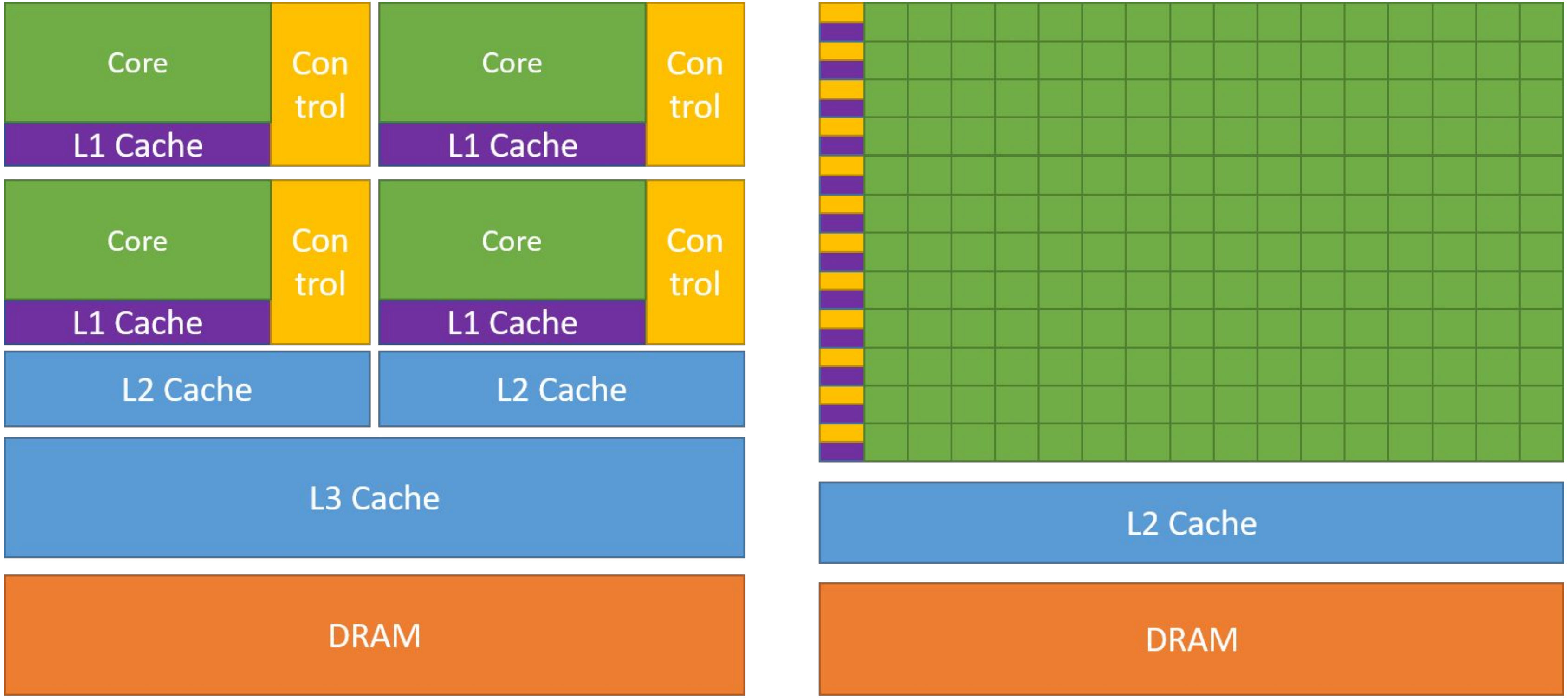
e.g. for many years (up to 2010s) this concept was presented in may motherboards by two chips:

[Northbridge] (CPU, GPU and memory)

[Southbridge] (all IO: PCI, SATA, Ethernet, USB)



CPU vs GPU



https://docs.nvidia.com/cuda/cuda-c-programming-guide/

Graphical Processing Unit

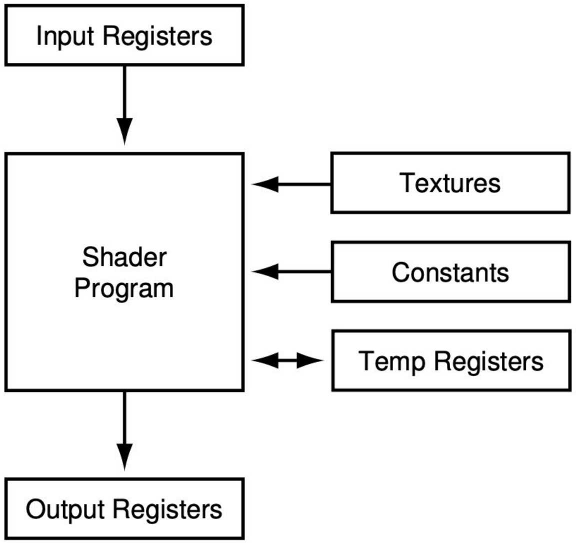


Figure 1: Programming model for current programmable graphics hardware. A shader program operates on a single input element (vertex or fragment) stored in the input registers and writes the execution result into the output registers.

Brook for GPUs: Stream Computing on Graphics Hardware, 2004

https://doi.org/10.1145/1015706.1015800

https://blogs.nvidia.com/blog/author/ian-buck/

CPU vs GPU

Types of Nvidia GPU

GPU Computing Applications

Libraries and Middleware

cuDNN TensorRT	cuFFT cuBLAS cuRAND cuSPARSE	CULA MAGMA	Thrust NPP	VSIPL SVM OpenCurrent	PhysX OptiX iRay	MATLAB Mathematica

Programming Languages

C	C++	Fortran	Java Python Wrappers	DirectCompute	Directives (e.g. OpenACC)

CUDA-Enabled NVIDIA GPUs

NVIDIA Ampere Architecture (compute capabilities 8.x)				Tesla A Series
NVIDIA Turing Architecture (compute capabilities 7.x)		GeForce 2000 Series	Quadro RTX Series	Tesla T Series
NVIDIA Volta Architecture (compute capabilities 7.x)	DRIVE/JETSON AGX Xavier		Quadro GV Series	Tesla V Series
NVIDIA Pascal Architecture (compute capabilities 6.x)	Tegra X2	GeForce 1000 Series	Quadro P Series	Tesla P Series

H100 hardware showcase

https://www.youtube.com/watch?v=3STJyioJr7k

Ampere / 30-Series

	GeForce RTX 3090 Ti	GeForce RTX 3090	GeForce RTX 3080 Ti	GeForce RTX 3080	GeForce RTX 3070 Ti	GeForce RTX 3070	GeForce RTX 3060 Ti	GeForce RTX 3060	GeForce RTX 3050
NVIDIA CUDA Cores	10752	10496	10240	8960 / 8704	6144	5888	4864	3584	2560 / 2304
Boost Clock (GHz)	1.86	1.70	1.67	1.71	1.77	1.73	1.67	1.78	1.78 / 1.76
Memory Size	24 GB	24 GB	12 GB	12 GB / 10 GB	8 GB	8 GB	8 GB	12 GB	8 GB
Memory Type	GDDR6X	GDDR6X	GDDR6X	GDDR6X	GDDR6X	GDDR6	GDDR6	GDDR6	GDDR6

Ada Lovelace / 40-Series

	GeForce RTX 4090	GeForce RTX 4080	GeForce RTX 4070 Ti
NVIDIA CUDA Cores	16384	9728	7680
Boost Clock (GHz)	2.52	2.51	2.61
Memory Size	24 GB	16 GB	12 GB
Memory Type	GDDR6X	GDDR6X	GDDR6X
Max Display Resolution	4K at 240Hz or 8K at 60Hz, with DSC	4K at 240Hz or 8K at 60Hz, with DSC	4K at 240Hz or 8K at 60Hz with DSC, HDR

How to choose a GPU

- choosing GPU https://www.youtube.com/watch?v=F1ythHjdWI0
- colab pro overview https://www.youtube.com/watch?v=ltnX0KDTfJQ
- one more choose GPU https://www.youtube.com/watch?v=YiX9p8A7LqE
    - https://docs.google.com/presentation/d/1VZBIqrtOoR2t1Fr-sIGJNeMNhLmKvcQD/edit#slide=id.p15
    - https://docs.google.com/spreadsheets/d/1mD4DrRcq3-y9r86O_P2fD7_0m2f0q9QI/edit?gid=542277665#gid=542277665
-

CUDA

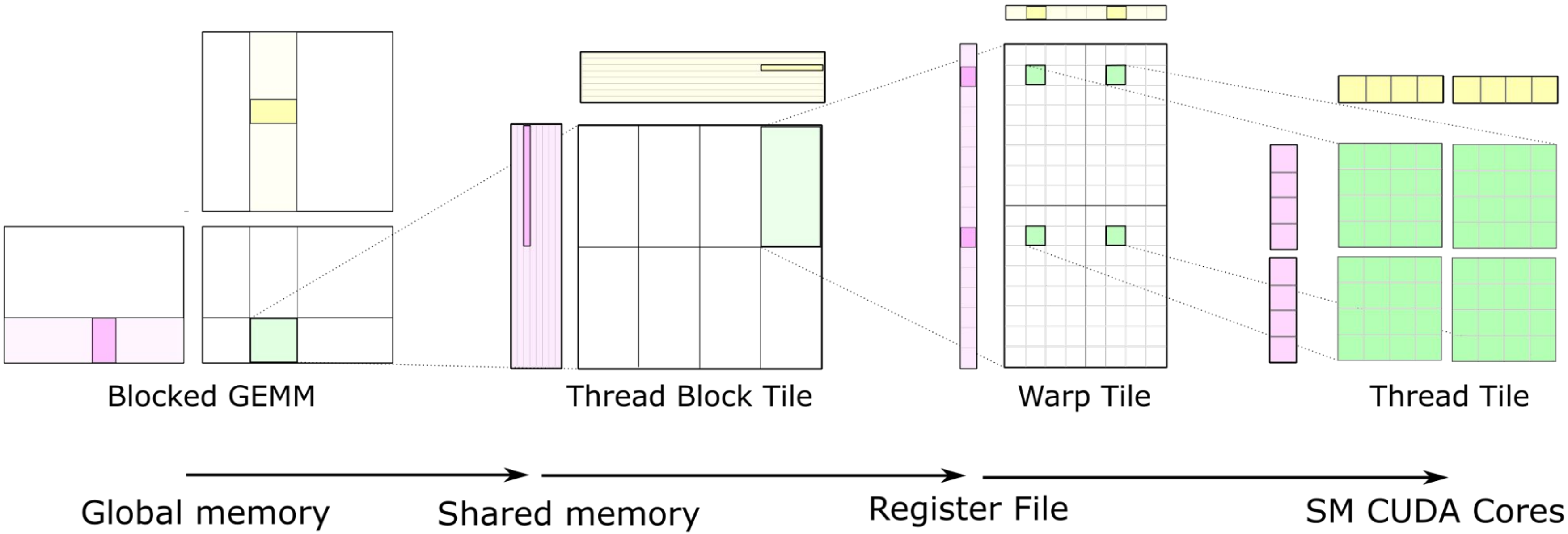
girafe
ai

02

Matrix multiplication

https://docs.nvidia.com/deeplearning/performance/dl-performance-matrix-multiplication/index.html

Block matrix multiplication



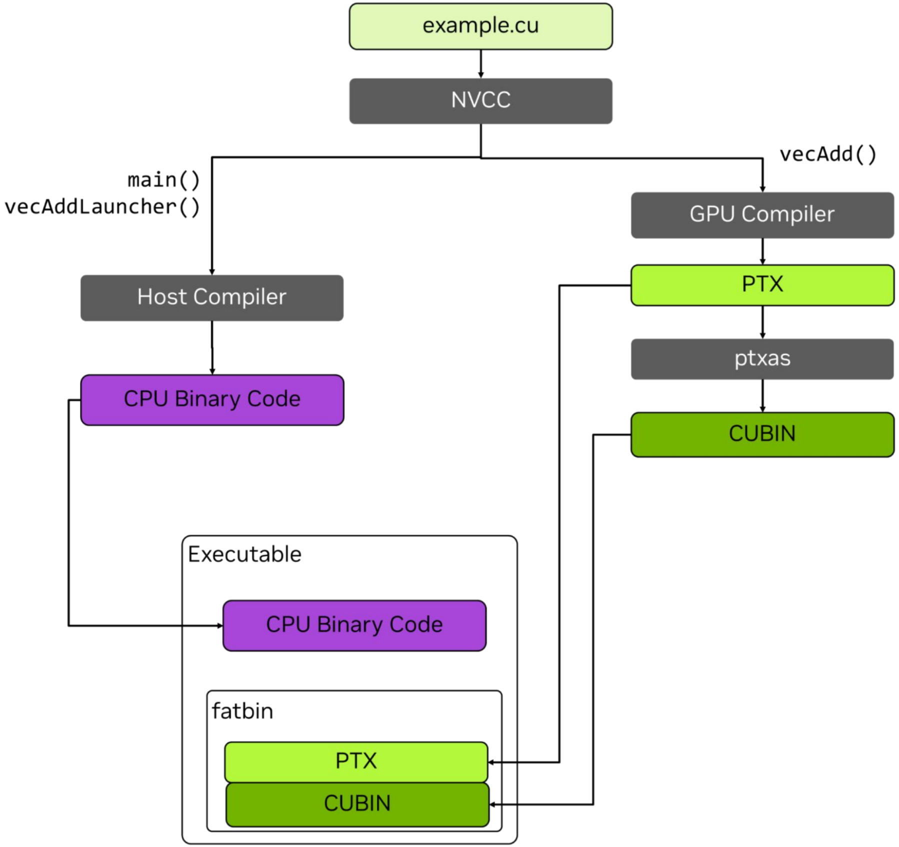
https://developer.nvidia.com/blog/cutlass-linear-algebra-cuda/

Deep Learning performance

https://docs.nvidia.com/deeplearning/performance/index.html

CUDA compilation

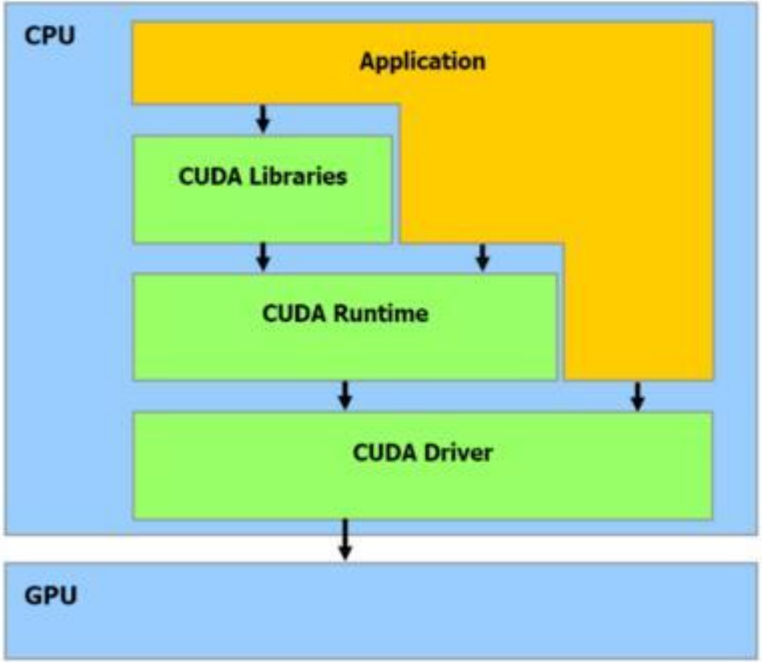
https://developer.nvidia.com/blog/understanding-ptx-the-assembly-language-of-cuda-gpu-computing/



CUDA software suite

- Driver
- Runtime
- Libraries
  - cuDNN
  - cuBLAS

more info



CUDA in Docker

Use official ones!

https://catalog.ngc.nvidia.com/orgs/nvidia/containers/cuda

Mixed precision

- https://docs.nvidia.com/deeplearning/performance/mixed-precision-training/index.html
-

More links

- Choose GPU
    - https://www.youtube.com/watch?v=H3AQnIpxk0c&t=107s
    - https://www.youtube.com/watch?v=ykDuoq-MpHg
-

Revise

1. Problem statement
2. Metrics
3. Models

Thanks for attention!

Questions?

girafe
ai